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- Get the textbook

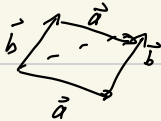
↳ He will follow the textbook

Vector operations

- magnitude & direction

$\vec{a}$

Addition:



$$\vec{a} + \vec{b} = \vec{b} + \vec{a}$$

(commutative)

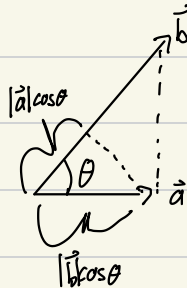
$$\vec{a} + (\vec{b} + \vec{c}) = (\vec{a} + \vec{b}) + \vec{c}$$

(associative)

Product

- Scalar/dot product

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$$



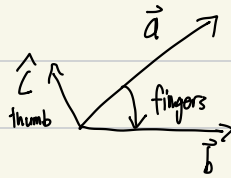
$$\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$$

- Vector/cross product

$$\vec{a} \times \vec{b} = |\vec{a}| |\vec{b}| \sin \theta \hat{c}$$

$\hat{c}$ is a unit vector normal to $\vec{a}$ & $\vec{b}$

• Right hand rule



Express a vector in a frame

Frames of reference:

• Includes:

↳ origin

↳ basis vectors

• must be orthogonal

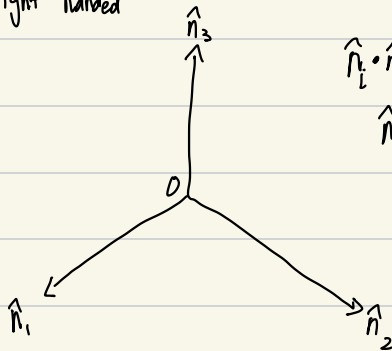
• unit magnitude

• right handed

Ref frame $\mathcal{N}: \{0, \hat{n}_1, \hat{n}_2, \hat{n}_3\}$

$$\hat{n}_i \cdot \hat{n}_j = 0 \text{ if } i \neq j$$

$$\hat{n}_i \cdot \hat{n}_i = 1 \text{ or } |\hat{n}_i| = 1$$

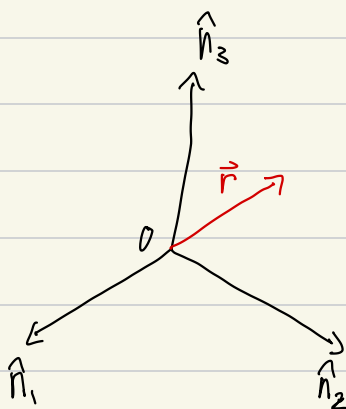


$$\hat{n}_1 \times \hat{n}_2 = \hat{n}_3$$

$$\hat{n}_2 \times \hat{n}_3 = \hat{n}_1$$

$$\hat{n}_3 \times \hat{n}_1 = \hat{n}_2$$

Express



$$\vec{r} = x\hat{n}_1 + y\hat{n}_2 + z\hat{n}_3$$

$$x = \vec{r} \cdot \hat{n}_1$$

$$y = \vec{r} \cdot \hat{n}_2$$

$$z = \vec{r} \cdot \hat{n}_3$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} \longleftrightarrow \vec{r} = r_{(N)} \quad \text{Vector } r \text{ expressed in the } N \text{ frame}$$

Revisit dot product

$$\vec{a} \rightarrow a_{(N)} = \begin{bmatrix} a_x \\ a_y \\ a_z \end{bmatrix}, \quad \vec{b} \rightarrow b_{(N)} = \begin{bmatrix} b_x \\ b_y \\ b_z \end{bmatrix}$$

$$\vec{a} \cdot \vec{b} = a_x b_x + a_y b_y + a_z b_z = a_{(N)}^T b_{(N)}$$

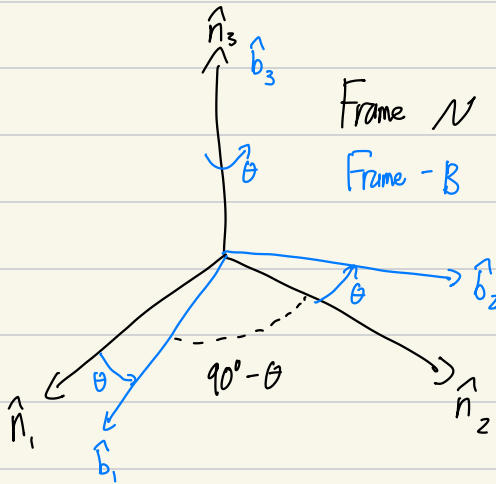
Revisit cross product

$$(\vec{a} \times \vec{b})_{(N)} = \tilde{a}_{(N)} b_{(N)} = \begin{bmatrix} 0 & -a_z & a_y \\ a_z & 0 & -a_x \\ -a_y & a_x & 0 \end{bmatrix} \begin{bmatrix} b_x \\ b_y \\ b_z \end{bmatrix} = \begin{bmatrix} a_y b_z - a_z b_y \\ a_z b_x - a_x b_z \\ a_x b_y - a_y b_x \end{bmatrix}$$

$$\text{or } \vec{a} \times \vec{b} = \begin{vmatrix} \hat{n}_1 & \hat{n}_2 & \hat{n}_3 \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix} \xrightarrow{\text{determinant}} = (a_y b_z - a_z b_y)\hat{n}_1 + (a_z b_x - a_x b_z)\hat{n}_2 + (a_x b_y - a_y b_x)\hat{n}_3$$

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Look into hours option



$$\hat{b}_1 = (\hat{b}_1 \cdot \hat{n}_1) \hat{n}_1 + (\hat{b}_1 \cdot \hat{n}_2) \hat{n}_2 + (\hat{b}_1 \cdot \hat{n}_3) \hat{n}_3$$

$$\hat{b}_1 \cdot \hat{n}_1 = |\hat{b}_1| |\hat{n}_1| \cos \theta = \cos \theta$$

$$\hat{b}_1 \cdot \hat{n}_2 = \cos(90^\circ - \theta) = \sin \theta \quad \hat{b}_1 \cdot \hat{n}_3 = 0$$

Do all combinations:

$$\begin{bmatrix} \hat{b}_1 \\ \hat{b}_2 \\ \hat{b}_3 \end{bmatrix} = \begin{bmatrix} \hat{b}_1 \cdot \hat{n}_1 & \hat{b}_1 \cdot \hat{n}_2 & \hat{b}_1 \cdot \hat{n}_3 \\ \hat{b}_2 \cdot \hat{n}_1 & \hat{b}_2 \cdot \hat{n}_2 & \hat{b}_2 \cdot \hat{n}_3 \\ \hat{b}_3 \cdot \hat{n}_1 & \hat{b}_3 \cdot \hat{n}_2 & \hat{b}_3 \cdot \hat{n}_3 \end{bmatrix} \begin{bmatrix} \hat{n}_1 \\ \hat{n}_2 \\ \hat{n}_3 \end{bmatrix} = \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \hat{n}_1 \\ \hat{n}_2 \\ \hat{n}_3 \end{bmatrix}$$

$$\begin{bmatrix} \hat{b}_1 \\ \hat{b}_2 \\ \hat{b}_3 \end{bmatrix} = C_{BN} \begin{bmatrix} \hat{n}_1 \\ \hat{n}_2 \\ \hat{n}_3 \end{bmatrix} \quad \text{direction cosine matrix}$$

$$\begin{bmatrix} \hat{n}_1 \\ \hat{n}_2 \\ \hat{n}_3 \end{bmatrix} = \begin{bmatrix} \hat{n}_1 \cdot \hat{b}_1 & \hat{n}_1 \cdot \hat{b}_2 & \hat{n}_1 \cdot \hat{b}_3 \\ \hat{n}_2 \cdot \hat{b}_1 & \hat{n}_2 \cdot \hat{b}_2 & \hat{n}_2 \cdot \hat{b}_3 \\ \hat{n}_3 \cdot \hat{b}_1 & \hat{n}_3 \cdot \hat{b}_2 & \hat{n}_3 \cdot \hat{b}_3 \end{bmatrix} \begin{bmatrix} \hat{b}_1 \\ \hat{b}_2 \\ \hat{b}_3 \end{bmatrix}$$

$$\begin{bmatrix} \hat{n}_1 \\ \hat{n}_2 \\ \hat{n}_3 \end{bmatrix} = C_{NB} \begin{bmatrix} \hat{b}_1 \\ \hat{b}_2 \\ \hat{b}_3 \end{bmatrix} = C_{BN}^T \begin{bmatrix} \hat{b}_1 \\ \hat{b}_2 \\ \hat{b}_3 \end{bmatrix}$$

$$\begin{bmatrix} \hat{n}_1 \\ \hat{n}_2 \\ \hat{n}_3 \end{bmatrix} = C_{NB} \begin{bmatrix} \hat{b}_1 \\ \hat{b}_2 \\ \hat{b}_3 \end{bmatrix} = \underbrace{C_{NB} C_{BN}}_I \begin{bmatrix} \hat{n}_1 \\ \hat{n}_2 \\ \hat{n}_3 \end{bmatrix}, \quad \begin{matrix} C_{NB} = C_{BN}^{-1} = \\ C_{BN}^T \end{matrix}$$

Properties of a DCM

- $C^T = C^{-1}$ "orthogonal matrix"
- Rows are orthogonal to each other
- Columns are orthogonal to each other
- row/col unit magnitude
- Can be used to convert a vector expressed in one frame to another

$$\vec{r} = r_{1n} \hat{n}_1 + r_{2n} \hat{n}_2 + r_{3n} \hat{n}_3 \quad \vec{r} \longleftrightarrow r_{(n)} = \begin{bmatrix} r_{1n} \\ r_{2n} \\ r_{3n} \end{bmatrix}$$

$$\text{find } C_{(B)} \quad \vec{r} = r_{1n} (C_{11} \hat{b}_1 + C_{12} \hat{b}_2 + C_{13} \hat{b}_3) + r_{2n} (C_{21} \hat{b}_1 + C_{22} \hat{b}_2 + C_{23} \hat{b}_3) \\ + r_{3n} (C_{31} \hat{b}_1 + C_{32} \hat{b}_2 + C_{33} \hat{b}_3)$$

Where $C_{NB} = \begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix}$

$$r_{(B)} = \begin{bmatrix} C_{11} r_{1n} + C_{21} r_{2n} + C_{31} r_{3n} \\ C_{12} r_{1n} + C_{22} r_{2n} + C_{32} r_{3n} \\ C_{13} r_{1n} + C_{32} r_{2n} + C_{33} r_{3n} \end{bmatrix}$$

$$= \begin{bmatrix} C_{11} & C_{21} & C_{31} \\ C_{12} & C_{22} & C_{32} \\ C_{13} & C_{23} & C_{33} \end{bmatrix} \begin{bmatrix} r_{1n} \\ r_{2n} \\ r_{3n} \end{bmatrix} \quad r_{(B)} = C_{NB}^T r_{(N)}$$

$$r_{(B)} = C_{BN} r_{(N)}$$

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$$r_{(B)} = \begin{bmatrix} C_{11}r_{1n} + C_{21}r_{2n} + C_{31}r_{3n} \\ C_{12}r_{1n} + C_{22}r_{2n} + C_{32}r_{3n} \\ C_{13}r_{1n} + C_{23}r_{2n} + C_{33}r_{3n} \end{bmatrix} \quad \therefore r_{(B)} = C_{Bn} r_{(n)}$$

Time derivative of a vector

$$\vec{r} = r_{1b} \hat{b}_1 + r_{2b} \hat{b}_2 + r_{3b} \hat{b}_3$$

$$\frac{d}{dt} \vec{r} = \left(\dot{r}_{1b} \hat{b}_1 + r_{1b} \frac{d}{dt} \hat{b}_1 \right) + \left(\dot{r}_{2b} \hat{b}_2 + r_{2b} \frac{d}{dt} \hat{b}_2 \right) + \left(\dot{r}_{3b} \hat{b}_3 + r_{3b} \frac{d}{dt} \hat{b}_3 \right)$$

a time derivative of a vector depends on your frame of reference

Concept: Specify which frame we are taking a derivative

$$^B \frac{d}{dt} \vec{r} = \dot{r}_{1b} \hat{b}_1 + \dot{r}_{2b} \hat{b}_2 + \dot{r}_{3b} \hat{b}_3$$

$$^N \frac{d}{dt} \vec{r} = \dot{r}_{1b} \hat{b}_1 + \dot{r}_{2b} \hat{b}_2 + \dot{r}_{3b} \hat{b}_3 + r_{1b} \frac{^N d}{dt} \hat{b}_1 + r_{2b} \frac{^N d}{dt} \hat{b}_2 + r_{3b} \frac{^N d}{dt} \hat{b}_3$$

Angular velocity:

$$\omega = \frac{d\theta}{dt}$$

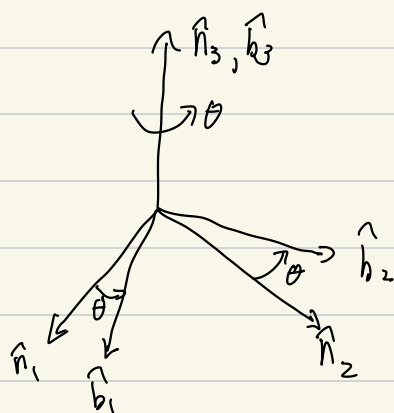
Angular velocity vector:

$$\vec{\omega} = \frac{d\theta}{dt} \hat{n}_w$$

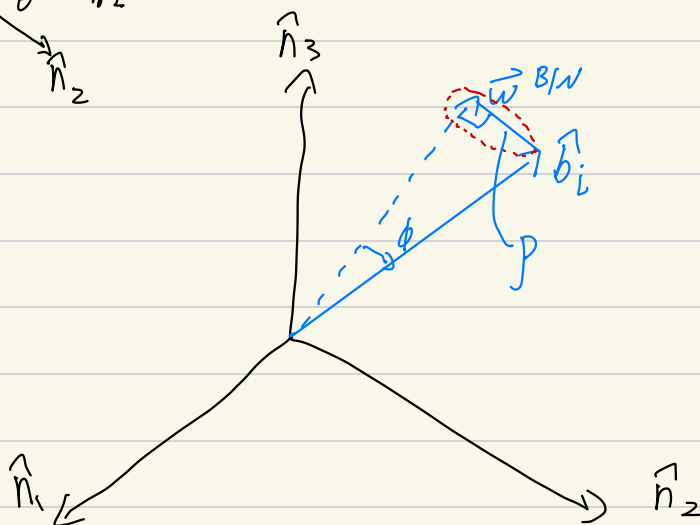
frame rotates relative to another by $\frac{d\theta}{dt}$ around unit vector $\hat{n}_w$

e.g. $\vec{\omega}^{B/N}$: angular velocity of frame B with respect to frame N

$$\vec{\omega}^{B/N} = \omega_{1n}^{B/N} \hat{n}_1 + \omega_{2n}^{B/N} \hat{n}_2 + \omega_{3n}^{B/N} \hat{n}_3$$



$$\vec{\omega}^{B/N} = \dot{\theta} \hat{n}_3 = \dot{\theta} \hat{b}_3$$



"r" "θ"

$$\left| \frac{d}{dt} \hat{b}_i \right| = \mathcal{P} \left| \vec{\omega}^{B/N} \right| = \sin \phi \left| \vec{\omega}^{B/N} \right|$$

direction of $\frac{d}{dt} \hat{b}_i$ perpendicular to $\hat{b}_i$ & $\vec{\omega}^{B/N}$

$$\frac{d}{dt} \hat{b}_i = \vec{\omega}^{B/N} \times \hat{b}_i \quad i=1,2,3$$

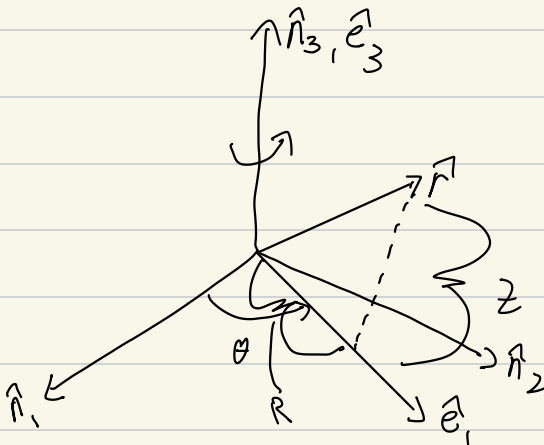
$$\frac{d}{dt} \vec{r} = \frac{d}{dt} \vec{r} + r_{1b} (\vec{\omega}^{B/N} \times \hat{b}_1) + r_{2b} (\vec{\omega}^{B/N} \times \hat{b}_2) + r_{3b} (\vec{\omega}^{B/N} \times \hat{b}_3)$$

$$\frac{d}{dt} \vec{r} = \frac{d}{dt} \vec{r} + \vec{\omega}^{B/N} \times (r_{1b} \hat{b}_1 + r_{2b} \hat{b}_2 + r_{3b} \hat{b}_3)$$

$\vec{r}$

$$\frac{d}{dt} \vec{r} = \frac{d}{dt} \vec{r} + \vec{\omega}^{B/N} \times \vec{r} \quad \text{transport thm.}$$

Ex)



$$\vec{r} = R \cos \theta \hat{n}_1 + R \sin \theta \hat{n}_2 + z \hat{n}_3$$

$$= R \hat{e}_1 + z \hat{e}_3$$

$$\frac{d}{dt} \vec{r} = (\dot{R} \cos \theta - \dot{\theta} R \sin \theta) \hat{n}_1 + (\dot{R} \sin \theta + \dot{\theta} R \cos \theta) \hat{n}_2 + \dot{z} \hat{n}_3$$

$$\begin{aligned} \frac{n}{d} \frac{d}{dt} \vec{r} &= \frac{B}{d} \frac{d}{dt} \vec{r} + \vec{\omega}^{B/N} \times \vec{r} = \dot{R} \hat{e}_1 + \dot{z} \hat{e}_3 \\ &\quad + \dot{\theta} \hat{e}_3 \times (R \hat{e}_1 + z \hat{e}_3) \\ &= \dot{R} \hat{e}_1 + \dot{\theta} R \hat{e}_2 + \dot{z} \hat{e}_3 \end{aligned}$$

$$\vec{\omega}^{B/N} = \dot{\theta} \hat{e}_3$$

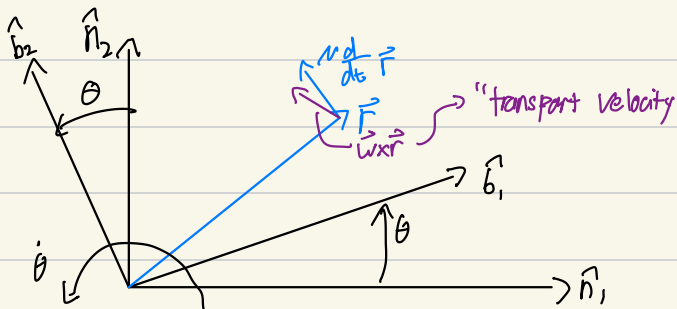
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$$\frac{n}{d} \frac{d}{dt} \vec{r} = \frac{n}{d} \dot{\vec{r}} \rightarrow \text{a vector}$$

$\dot{\vec{r}}_{(N)} = C_{NE} \dot{\vec{r}}_{(E)}$ If a derivative is taken in the same frame as it is expressed in, then it's just the derivative of the components:

If not, then $\vec{\omega} \times \vec{r}$ terms, transport thm:

$$\frac{n}{d} \frac{d}{dt} \vec{r} = \frac{B}{d} \frac{d}{dt} \vec{r} + \vec{\omega}^{B/N} \times \vec{r}$$



instantaneous movement of point fixed in frame-B where $\vec{r}$ is currently located to an observer in frame n

Acceleration

transport thm: $\overset{N}{\frac{d}{dt}} () = \overset{B}{\frac{d}{dt}} () + \vec{\omega}^{B/N} \times ()$

$$\vec{v} = \overset{N}{\frac{d}{dt}} \vec{r} = \overset{B}{\frac{d}{dt}} \vec{r} + \vec{\omega}^{B/N} \times \vec{r}$$

$$\overset{N}{\frac{d}{dt}} \left(\overset{N}{\frac{d}{dt}} \vec{r} \right) = \overset{B}{\frac{d}{dt}} \left(\overset{B}{\frac{d}{dt}} \vec{r} + \vec{\omega}^{B/N} \times \vec{r} \right) + \vec{\omega}^{B/N} \times \left(\overset{B}{\frac{d}{dt}} \vec{r} + \vec{\omega}^{B/N} \times \vec{r} \right)$$

$$\overset{N}{\frac{d^2}{dt^2}} \vec{r} = \overset{B}{\frac{d^2}{dt^2}} \vec{r} + \overset{B}{\frac{d}{dt}} \vec{\omega}^{B/N} \times \vec{r} + \vec{\omega}^{B/N} \times \overset{B}{\frac{d}{dt}} \vec{r} + \vec{\omega}^{B/N} \times \overset{B}{\frac{d}{dt}} \vec{r} +$$

$$\vec{\omega}^{B/N} \times (\vec{\omega}^{B/N} \times \vec{r})$$

$$\overset{N}{\ddot{\vec{r}}} = \overset{B}{\ddot{\vec{r}}} + \overset{B}{\dot{\vec{\omega}}}^{B/N} \times \vec{r} + \underbrace{2 \vec{\omega}^{B/N} \times \overset{B}{\dot{\vec{r}}}}_{\text{tangential acceleration}} + \underbrace{\vec{\omega}^{B/N} \times (\vec{\omega}^{B/N} \times \vec{r})}_{\text{Centrifugal / Centripetal}} + \underbrace{\vec{\omega}^{B/N} \times \overset{B}{\frac{d}{dt}} \vec{r}}_{\text{Coriolis}}$$

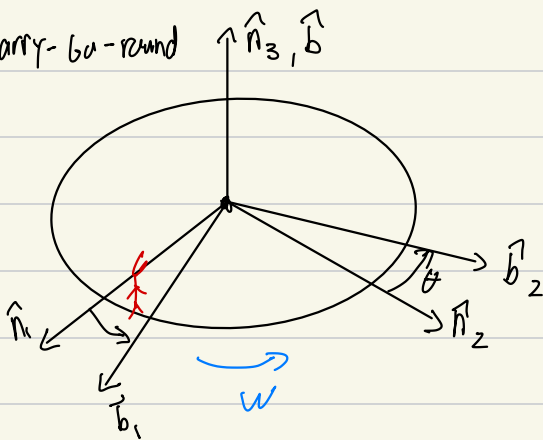
Inertial frame: a frame where $\vec{F} = m\vec{a}$

• "not rotating", "not accelerating"

If frame N is inertial, then

$$\overset{N}{\frac{d^2}{dt^2}} \vec{r} = \vec{a} = \frac{\vec{F}}{m} = \overset{B}{\ddot{\vec{r}}} + \overset{B}{\dot{\vec{\omega}}}^{B/N} \times \vec{r} + 2 \vec{\omega}^{B/N} \times \overset{B}{\dot{\vec{r}}} + \vec{\omega}^{B/N} \times (\vec{\omega}^{B/N} \times \vec{r})$$

Marry-go-round



$$\vec{\omega}^{B/N} = \dot{\theta} \hat{n}_3$$

Constant

Frame - B fixed to the
marry-go-round

$$\vec{r} = R \hat{b}_1, \quad R \text{ not constant}$$

$$\frac{\vec{F}}{m} = \ddot{R} \hat{b}_1 + 2\dot{\theta} \hat{b}_3 \times \dot{R} \hat{b}_1 + \dot{\theta} \hat{b}_3 \times (\dot{\theta} \hat{b}_3 \times R \hat{b}_1)$$

$$\frac{\vec{F}}{m} = \ddot{R} \hat{b}_1 + 2\dot{\theta} R \hat{b}_2 - \dot{\theta}^2 R \hat{b}_1$$

${}^B \vec{r} = R \hat{b}_1$

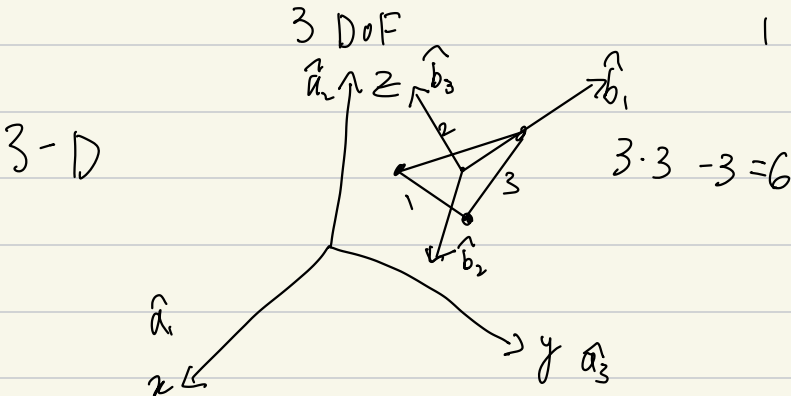
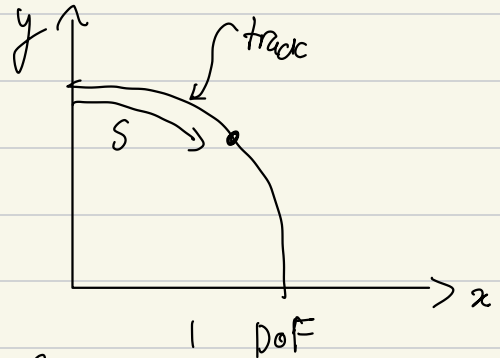
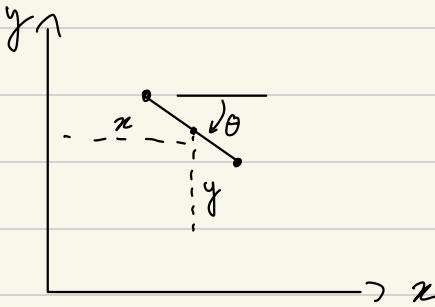
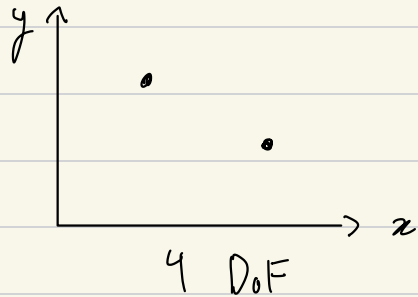
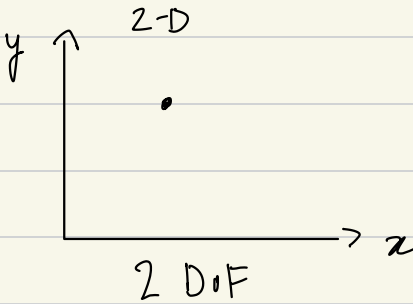
$$\text{if } \ddot{R} = 0, \dot{R} = 0 \rightarrow \frac{\vec{F}}{m} = -\dot{\theta}^2 R \hat{b}_1$$

$$\text{If } \dot{R} \neq 0 \quad \frac{\vec{F}}{m} = -\dot{\theta}^2 R \hat{b}_1 + 2\dot{\theta} R \hat{b}_2$$

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Degrees of freedom DoF

Minimum number of parameters to describe the configuration of a system (dimensions times # of particles - # of constraints)



Specify orientation

$$\begin{bmatrix} \hat{b}_1 \\ \hat{b}_2 \\ \hat{b}_3 \end{bmatrix} = \begin{bmatrix} L_{11} & L_{12} & L_{13} \\ L_{21} & L_{22} & L_{23} \\ L_{31} & L_{32} & L_{33} \end{bmatrix} \begin{bmatrix} \hat{a}_1 \\ \hat{a}_2 \\ \hat{a}_3 \end{bmatrix}$$

DCM has 9 elements

$$\hat{b}_1 \cdot \hat{b}_1 = 1 = (L_{11}\hat{a}_1 + L_{12}\hat{a}_2 + L_{13}\hat{a}_3) \cdot (L_{11}\hat{a}_1 + L_{12}\hat{a}_2 + L_{13}\hat{a}_3)$$

$$L_{11}^2 + L_{12}^2 + L_{13}^2 = 1$$

Magnitude of row is 1

$$\hat{b}_1 \cdot \hat{b}_2 = 0 = (L_{11}\hat{a}_1 + L_{12}\hat{a}_2 + L_{13}\hat{a}_3) \cdot (L_{21}\hat{a}_1 + L_{22}\hat{a}_2 + L_{23}\hat{a}_3)$$

$$L_{11}L_{21} + L_{12}L_{22} + L_{13}L_{23} = 0$$

$$\begin{array}{lll} \text{6 constraints:} & \hat{b}_1 \cdot \hat{b}_1 = 1 & \hat{b}_2 \cdot \hat{b}_2 = 1 & \hat{b}_3 \cdot \hat{b}_3 = 1 \\ & \hat{b}_1 \cdot \hat{b}_2 = 0 & \hat{b}_1 \cdot \hat{b}_3 = 0 & \hat{b}_2 \cdot \hat{b}_3 = 0 \end{array}$$

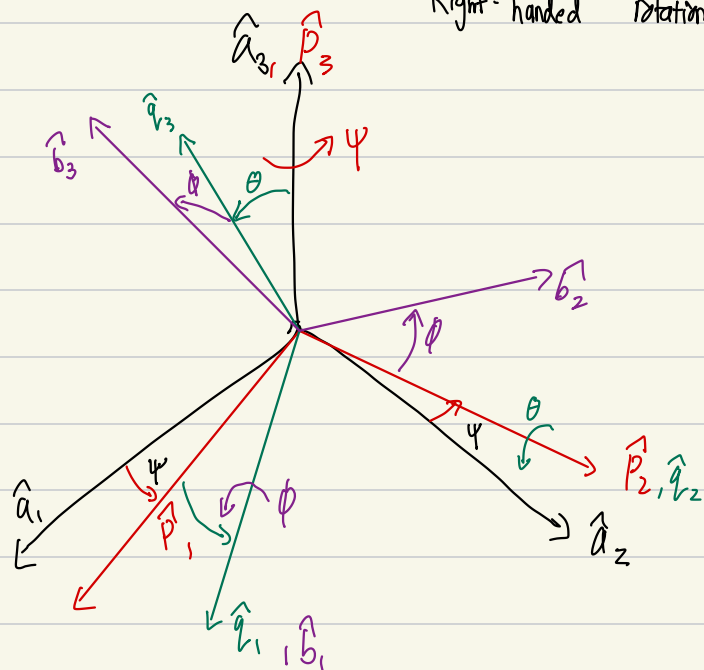
Euler: Any rotation/orientation can be represented by a sequence of 3 rotations around non-parallel basis Vectors

"Euler Angles" Specify order of rotations

Order of rotations: 3-2-1, 3-1-3

3-2-1 would: • Right-handed rotation about $\hat{a}_3$

- Right-handed rotation about new 2-axis
- Right-handed rotation about new 1-axis



$$L_{PA} = L_3(\psi) =$$

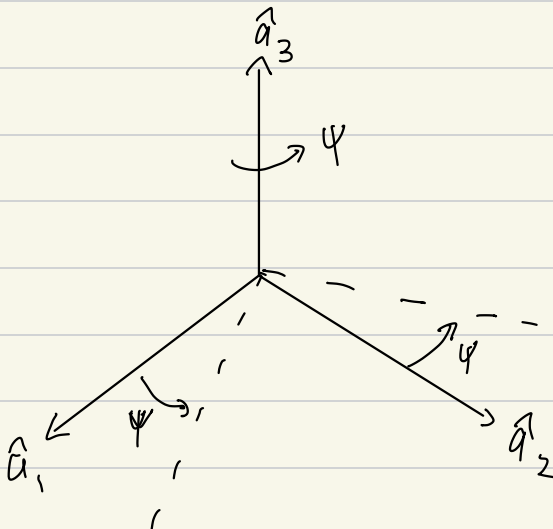
$$\begin{bmatrix} \cos \psi & \sin \psi & 0 \\ -\sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\vec{\omega}^{P/A} = \dot{\psi} \hat{p}_3 = \dot{\psi} \hat{a}_3$$

$$C_{QP} = C_2(\theta) = \begin{bmatrix} \cos \theta & 0 & -\sin \theta \\ 0 & 1 & 0 \\ \sin \theta & 0 & \cos \theta \end{bmatrix} \quad \vec{\omega}^{Q/P} = \dot{\theta} \hat{p}_2 = \dot{\theta} \hat{q}_2$$

$$C_{BA} = C_1(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \phi & \sin \phi \\ 0 & -\sin \phi & \cos \phi \end{bmatrix}$$

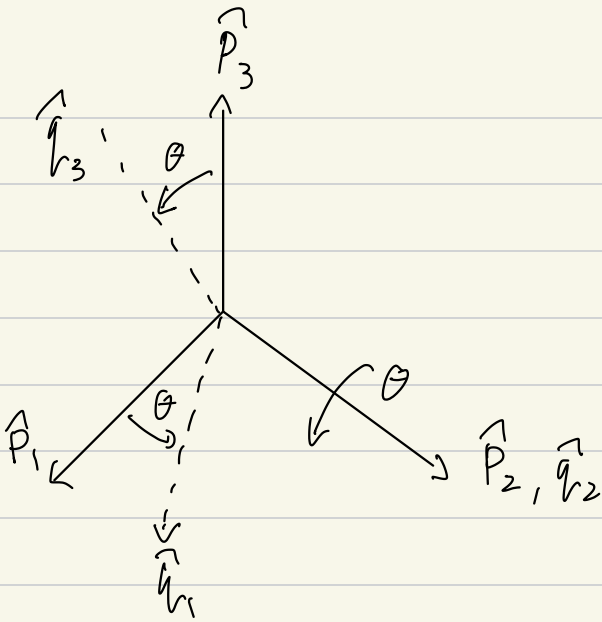
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$$\vec{\omega}^{P/A} = \dot{\psi} \hat{p}_3 = \dot{\psi} \hat{a}_3$$

$$C_{PA} = C_3(\psi)$$

$$\begin{bmatrix} \cos \psi & \sin \psi & 0 \\ -\sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



$$\vec{\omega}^{q/p} = \dot{\theta} \hat{p}_2 = \dot{\theta} \hat{q}_2$$

$$L_{qp} = L_2(\theta)$$

$$\begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix}$$

$$\begin{bmatrix} \hat{p}_1 \\ \hat{p}_2 \\ \hat{p}_3 \end{bmatrix} = L_{pq} \begin{bmatrix} \hat{q}_1 \\ \hat{q}_2 \\ \hat{q}_3 \end{bmatrix}$$

$$\begin{bmatrix} \hat{q}_1 \\ \hat{q}_2 \\ \hat{q}_3 \end{bmatrix} = L_{qp} \begin{bmatrix} \hat{p}_1 \\ \hat{p}_2 \\ \hat{p}_3 \end{bmatrix}$$

$$= L_{qp} L_{pq} \begin{bmatrix} \hat{q}_1 \\ \hat{q}_2 \\ \hat{q}_3 \end{bmatrix}$$

$$\begin{bmatrix} \hat{b}_1 \\ \hat{b}_2 \\ \hat{b}_3 \end{bmatrix} = L_{Bq} \begin{bmatrix} \hat{q}_1 \\ \hat{q}_2 \\ \hat{q}_3 \end{bmatrix} = L_{Bq} L_{qp} L_{pq} \begin{bmatrix} \hat{q}_1 \\ \hat{q}_2 \\ \hat{q}_3 \end{bmatrix}$$

$$r_{(B)} = \underbrace{L_{Bq} L_{qp} L_{pq}}_{\text{"321"}} r_{(A)} = L_{BA} r_{(A)}$$

equal to

$$r_{(B)} = L_1(\psi) L_2(\theta) L_3(\varphi) r_{(A)}$$

$$C_{BA} = C_1(\phi) C_2(\theta) C_3(\psi) \quad \phi, \theta, \psi \text{ are Euler angles}$$

How many combinations are possible? (Right handed rotations around intermediate basis vectors)

Rotation #1 #2 #3

Options 3 2 2 $3 \cdot 2 \cdot 2 = 12$ possibilities

$$\vec{w}^{B/A} = \vec{w}^{B/Q} + \vec{w}^{Q/P} + \vec{w}^{P/A} = \dot{\psi} \hat{p}_3 + \dot{\theta} \hat{q}_2 + \dot{\phi} \hat{b}_1$$

($\hat{p}_3, \hat{q}_2, \hat{b}_1$ aren't orthogonal vectors)

• Lets say we want $\vec{w}^{P/A}$ in frame B

we need $\hat{p}_3$ & $\hat{q}_2$ in the B frame

$$C_1(\phi) = C_{BA} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & \sin\phi \\ 0 & -\sin\phi & \cos\phi \end{bmatrix} \quad \begin{bmatrix} \hat{b}_1 \\ \hat{b}_2 \\ \hat{b}_3 \end{bmatrix} = C_1(\phi) \begin{bmatrix} \hat{q}_1 \\ \hat{q}_2 \\ \hat{q}_3 \end{bmatrix}$$

$$\hat{b}_2 = \cos\phi \hat{q}_2 + \sin\phi \hat{q}_3$$

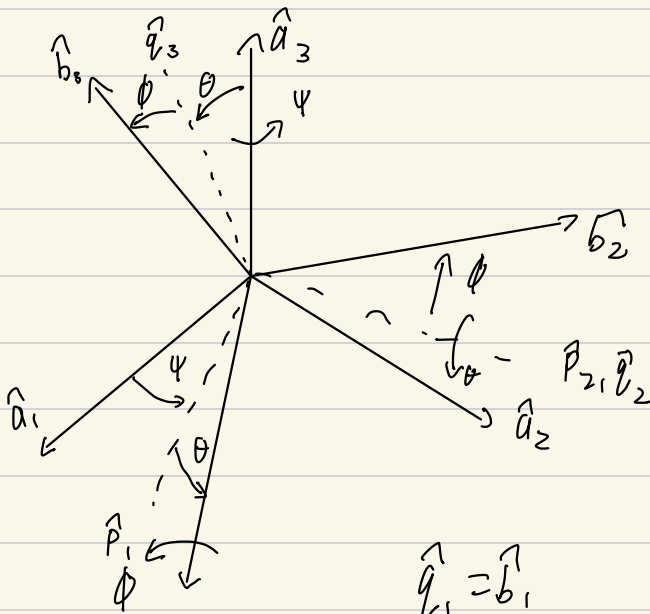
$$\hat{b}_3 = -\sin\phi \hat{q}_2 + \cos\phi \hat{q}_3$$

$$\hat{q}_2 = \cos\phi \hat{b}_2 - \sin\phi \hat{b}_3$$

$$\hat{q}_3 = \sin\phi \hat{b}_2 + \cos\phi \hat{b}_3$$

$$C_2(\theta) = C_{ap} = \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix}$$

q/q



$$\vec{w}^{B/A} = \vec{w}^{B/Q} + \vec{w}^{Q/P} + \vec{w}^{P/A}$$

$$\vec{w}^{B/A} = \dot{\psi} \hat{p}_3 + \dot{\theta} \hat{q}_2 + \dot{\phi} \hat{b}_1$$

$$\hat{q}_1 = \hat{b}_1$$

$$\hat{q}_2 = \cos\phi \hat{b}_2 - \sin\phi \hat{b}_3 \leftarrow \text{need}$$

$$\hat{q}_3 = \sin\phi \hat{b}_2 + \cos\phi \hat{b}_3$$

$$C_2(\theta) = C_{ap} = \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix}$$

$$\hat{q}_1 = \cos\theta \hat{p}_1 - \sin\theta \hat{p}_3$$

$$\hat{q}_2 = \hat{p}_2$$

$$\hat{q}_3 = \sin\theta \hat{p}_1 + \cos\theta \hat{p}_3$$

$$\hat{p}_1 = \cos\theta \hat{q}_1 + \sin\theta \hat{q}_3$$

$$\hat{p}_2 = \hat{q}_2$$

$$\hat{p}_3 = -\sin\theta \hat{q}_1 + \cos\theta \hat{q}_3 \leftarrow \text{need}$$

$$\hat{p}_3 = -\sin\theta \hat{b}_1 + \cos\theta (\sin\phi \hat{b}_2 + \cos\phi \hat{b}_3)$$

$$\vec{\omega}^{B/A} = \dot{\psi}(-\sin\theta \hat{b}_1 + \cos\theta (\sin\phi \hat{b}_2 + \cos\phi \hat{b}_3)) + \dot{\theta}(\cos\phi \hat{b}_2 - \sin\phi \hat{b}_3) + \dot{\phi} \hat{b}_1$$

$$= (-\dot{\psi} \sin\theta + \dot{\phi}) \hat{b}_1 + (\dot{\psi} \cos\theta \sin\phi + \dot{\theta} \cos\phi) \hat{b}_2 + (\dot{\psi} \cos\theta \cos\phi - \dot{\theta} \sin\phi) \hat{b}_3$$

$$\vec{\omega}_{(B)}^{B/A} = \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix} = \begin{bmatrix} \dot{\phi} - \dot{\psi} \sin\theta \\ \dot{\theta} \cos\phi + \dot{\psi} \cos\theta \sin\phi \\ -\dot{\theta} \sin\phi + \dot{\psi} \cos\theta \cos\phi \end{bmatrix}$$

$$\begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = \text{find}$$

$$\begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} = \underbrace{\begin{bmatrix} -\sin\theta & 0 & 1 \\ \cos\theta \sin\phi & \cos\phi & 0 \\ \cos\theta \cos\phi & -\sin\phi & 0 \end{bmatrix}}_{B^{-1}} \begin{bmatrix} \dot{\psi} \\ \dot{\theta} \\ \dot{\phi} \end{bmatrix}$$

B^{-1}

Note: $B^T \neq B^{-1}$, not a DCM

$$\begin{bmatrix} \dot{\psi} \\ \dot{\theta} \\ \dot{\phi} \end{bmatrix} = B \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} \quad B = \frac{1}{\cos\theta} \begin{bmatrix} 0 & \sin\phi & \cos\phi \\ 0 & \cos\theta \cos\phi & -\cos\theta \sin\phi \\ \cos\theta & \sin\theta \sin\phi & \sin\theta \cos\phi \end{bmatrix}$$

Singularity at $\theta = \pm 90^\circ$

- If the second rotation makes it so that the first and third rotations are around the same axis, we get a singularity
- the problem is ψ & ϕ are no longer unique

$$\phi, \theta, \psi \rightarrow \mathcal{L}_{BA}$$

$\mathcal{L}_{BA} \rightarrow \psi, \theta, \phi$ Unique solution doesn't necessarily exist

$$C_3(\psi) = \begin{bmatrix} \cos \psi & \sin \psi & 0 \\ -\sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$C_2(\theta) = \begin{bmatrix} \cos \theta & 0 & -\sin \theta \\ 0 & 1 & 0 \\ \sin \theta & 0 & \cos \theta \end{bmatrix}$$

$$C_1(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \phi & \sin \phi \\ 0 & -\sin \phi & \cos \phi \end{bmatrix}$$

$$C_{BA} = C_1(\phi) C_2(\theta) C_3(\psi)$$

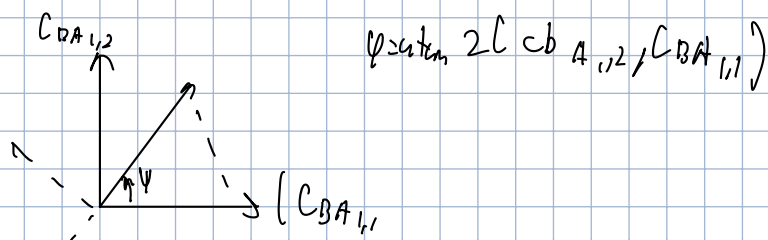
$$C_{BA} =$$

$$\begin{bmatrix} \cos \theta \cos \psi & \cos \theta \sin \psi & -\sin \theta \\ \sin \phi \sin \theta \cos \psi - \cos \phi \sin \psi & \sin \phi \sin \theta \sin \psi + \cos \phi \cos \psi & \sin \phi \cos \theta \\ \cos \phi \sin \theta \cos \psi + \sin \phi \sin \psi & \cos \phi \sin \theta \sin \psi - \sin \phi \cos \psi & \cos \phi \cos \theta \end{bmatrix}$$

$$C_{BA,1,3} = -\sin \theta$$

$$\theta = -\sin^{-1}(C_{BA,1,3}) = -\arcsin(C_{BA,1,3})$$

$$\frac{C_{BA,1,2}}{C_{BA,1,1}} = \frac{\cancel{\cos \theta} \sin \psi}{\cancel{\cos \theta} \cos \psi} = \tan^{-1} \left(\frac{C_{BA,1,2}}{C_{BA,1,1}} \right) = \psi$$



$$\arctan(s, c) = \psi$$

$$= \begin{cases} \arctan\left(\frac{s}{c}\right) & c > 0 \\ \arctan\left(\frac{s}{c}\right) + \pi & \text{otherwise} \end{cases}$$

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$$\theta = -\sin^{-1}(C_{BA1,3})$$

$$\psi = \alpha \tan 2(C_{BA1,2}, C_{BA1,1})$$

$$\phi = \alpha \tan 2(C_{BA2,3}, C_{BA3,3})$$

$$C_{BA} = \begin{bmatrix} \cos\theta \cos\psi & \cos\theta \sin\psi & -\sin\theta \\ \sin\phi \sin\theta \cos\psi - \cos\phi \sin\psi & \sin\phi \sin\theta \sin\psi + \cos\phi \cos\psi & \sin\phi \cos\theta \\ \cos\phi \sin\theta \cos\psi + \sin\phi \sin\psi & \cos\phi \sin\theta \sin\psi - \sin\phi \cos\psi & \cos\phi \cos\theta \end{bmatrix}$$

$$\rightarrow \tan^{-1}\left(\frac{C_{BA1,2}}{C_{BA1,1}}\right) \quad \theta = \pm 90^\circ \text{ Singularity}$$

$$\theta = 90^\circ \quad C_{BA} = \begin{bmatrix} 0 & 0 & -1 \\ \sin\phi \cos\psi - \cos\phi \sin\psi & \sin\phi \sin\psi + \cos\phi \cos\psi & 0 \\ \cos\phi \cos\psi + \sin\phi \sin\psi & \cos\phi \sin\psi - \sin\phi \cos\psi & 0 \end{bmatrix}$$

$$C_{BA} = \begin{bmatrix} 0 & 0 & -1 \\ \sin(\phi-\psi) & \cos(\phi-\psi) & 0 \\ \cos(\phi-\psi) & -\sin(\phi-\psi) & 0 \end{bmatrix}$$

Still $\theta = 90^\circ$

$$\begin{matrix} & B^{-1} \\ 3-2-1 \end{matrix} \quad \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} = \begin{bmatrix} -\sin\theta & 0 & 1 \\ \cos\theta \sin\phi & \cos\phi & 0 \\ \cos\theta \cos\phi & -\sin\phi & 0 \end{bmatrix} \begin{bmatrix} \dot{\psi} \\ \dot{\theta} \\ \dot{\phi} \end{bmatrix} \xrightarrow{\theta=90^\circ} \begin{bmatrix} -1 & 0 & 1 \\ 0 & \cos\phi & 1 \\ 0 & \sin\phi & 0 \end{bmatrix} \begin{bmatrix} \dot{\psi} \\ \dot{\theta} \\ \dot{\phi} \end{bmatrix}$$

$$w_1 = -\dot{\phi} + \dot{\psi} = -\frac{d}{dt}(\phi - \psi)$$

$$w_2 = \dot{\theta} \cos\phi$$

$$w_3 = -\dot{\theta} \sin\phi$$

Consider 3-1-3 Sequence

$\psi \rightarrow \theta_1$ 1st rotation 3rd axis

$\theta \rightarrow \theta_2$ 2nd rotation 1st axis

$\phi \rightarrow \theta_3$ 3rd rotation 3rd axis

$$B = \frac{1}{\sin \theta_2} \begin{bmatrix} \sin \theta_3 & \cos \theta_3 & 0 \\ \sin \theta_2 \cos \theta_3 & -\sin \theta_2 \sin \theta_3 & 0 \\ -\cos \theta_2 \sin \theta_3 & -\cos \theta_2 \cos \theta_3 & \sin \theta_2 \end{bmatrix}$$

$$\begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \dot{\theta}_3 \end{bmatrix} = B \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix}$$

Symmetric Sets have a Singularity at $\theta_2 = 0$ or 180°
 e.g. 3-1-3, 2-1-2, 1-2-1, etc

Asymmetric Sets have singularity at $\theta_2 = \pm 90^\circ$
 e.g. 3-2-1, 2-1-3, 1-2-3, etc

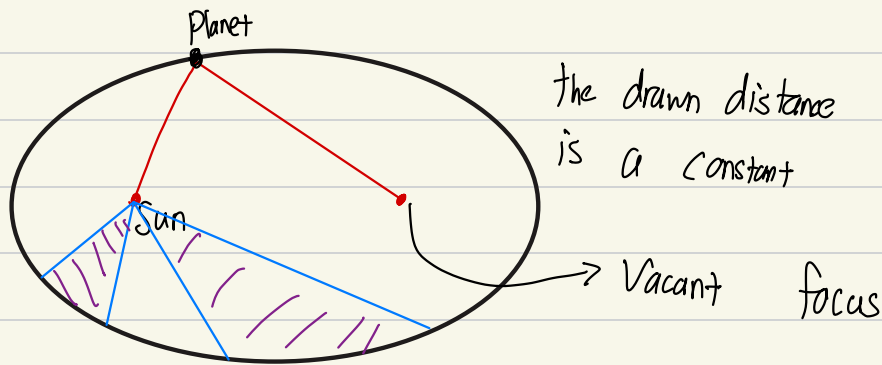
- Always have two singularities in opposite directions

- Sometimes things are conserved in certain directions (e.g. angular momentum)
 Choose so that Euler angles become constant (some of them)

9/18 Kepler's Laws

Johannes Kepler: 1571-1630

1st law (1609): The orbit of a planet is an ellipse, with the Sun located at one of its foci



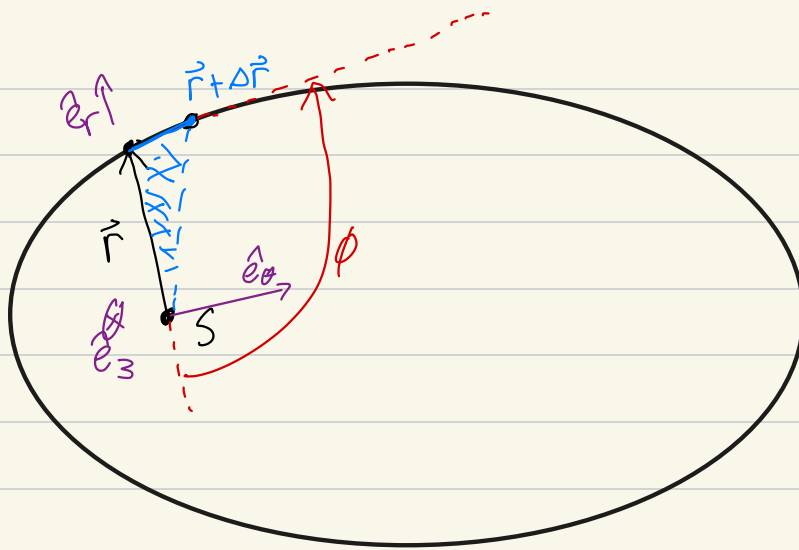
2nd law (1609): The line joining the planet and the Sun sweeps out equal areas in equal amounts of time

3rd law (1619): The square of a planet's orbital period (year) is proportional to the cube of the planet's mean distance from the Sun

$$T^2 \sim R^3$$

length of year $\rightarrow$ T $\rightarrow$ average radius R

Newton (1643-1727) will help us understand why these laws are valid - and explain even more



$$\text{Swept area} = \frac{1}{2} |\vec{r}| |\Delta \vec{r}| \sin \phi$$

$$(\text{Area}) \hat{e}_3 = \frac{1}{2} \vec{r} \times \Delta \vec{r} \quad \frac{1}{2} \vec{r} \times \Delta \vec{r} = \hat{e}_3 \Delta \text{area} = \text{const}$$

divide by Δt

$$\frac{1}{2} \vec{r} \times \frac{\Delta \vec{r}}{\Delta t} = \hat{e}_3 \frac{\Delta \text{area}}{\Delta t} \rightarrow \frac{1}{2} (\vec{r} \times \vec{v}) = \hat{e}_3 \frac{d}{dt} \text{Area} \quad ; \quad \vec{r} \times \vec{v} = \text{const}$$

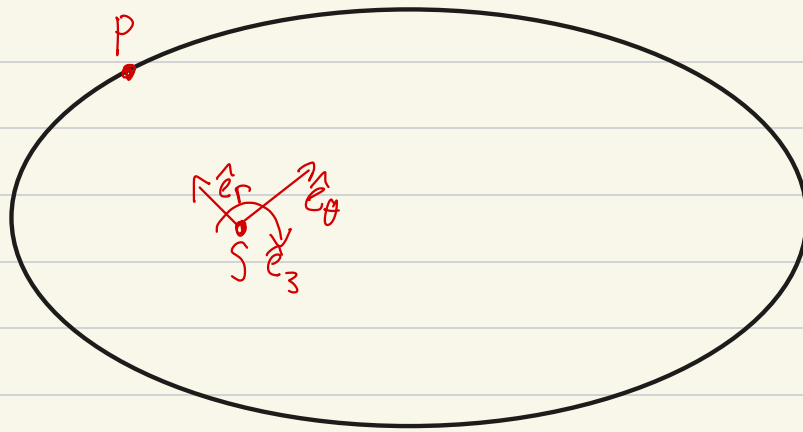
Define ref fram $E\{\hat{e}_r, \hat{e}_\theta, \hat{e}_3\}$ $\vec{r} = r \hat{e}_3 \quad ; \quad r = |\vec{r}|$

$$\vec{v} = \dot{\vec{r}} = \dot{r} \hat{e}_r + r \dot{\hat{e}}_r = \dot{r} \hat{e}_r + r (\vec{\omega}^{E/W} \times \hat{e}_r)$$

$$\vec{v} = \dot{r} \hat{e}_r + r \dot{\theta} \hat{e}_\theta$$

$$\vec{\omega}^{E/W} = \dot{\theta} \hat{e}_3$$

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$$\vec{r} = r\hat{e}_r$$

$$\vec{v} = \dot{r}\hat{e}_r + r\dot{\theta}\hat{e}_\theta$$

$$\vec{r} \times \vec{v} \text{ conserved}, \quad \vec{r} \times \vec{v} = r\hat{e}_r \times (\dot{r}\hat{e}_r + r\dot{\theta}\hat{e}_\theta) = r^2\dot{\theta}\hat{e}_3$$

$\vec{h} = \vec{r} \times \vec{v}$ is specific angular momentum; Specific = per unit mass

$$\text{Angular Momentum } \vec{H} = \vec{r} \times (m\vec{v}), \quad \vec{h} = \frac{\vec{H}}{m} = \vec{r} \times \vec{v}$$

What causes $\vec{H}$ to change

$$\frac{d\vec{H}}{dt} = \vec{v} \times m\vec{v} + \vec{r} \times m\vec{a}$$

$$\frac{d\vec{H}}{dt} = \vec{r} \times \vec{F} = 0 \quad \text{for an orbit}$$

$$r\hat{e}_r \times \vec{F} = 0$$

F is parallel to $\hat{e}_r$
force acting on the planet are radial

$$\vec{a} = \frac{d}{dt} \vec{v} = \frac{d}{dt} (\dot{r} \hat{e}_r + r \dot{\theta} \hat{e}_\theta) = \ddot{r} \hat{e}_r + \dot{r} \dot{\hat{e}}_r + \dot{r} \dot{\theta} \hat{e}_\theta + r \ddot{\theta} \hat{e}_\theta + r \dot{\theta} \dot{\hat{e}}_\theta \rightarrow$$

$$\dot{\hat{e}}_r = \vec{\omega}^{E/N} \times \hat{e}_r = \dot{\theta} \hat{e}_3 \times \hat{e}_r = \dot{\theta} \hat{e}_\theta$$

$$\dot{\hat{e}}_\theta = \vec{\omega}^{E/N} \times \hat{e}_\theta = \dot{\theta} \hat{e}_3 \times \hat{e}_\theta = -\dot{\theta} \hat{e}_r$$

$$\vec{a} = (\ddot{r} - r\dot{\theta}^2) \hat{e}_r + (2\dot{r}\dot{\theta} + r\ddot{\theta}) \hat{e}_\theta \quad \text{---} \quad \vec{a} = 0$$

$$\frac{d}{dt} (r^2 \dot{\theta}) = 0 \rightarrow \text{we know}$$

$$\underbrace{(2\dot{r}\dot{\theta} + r\ddot{\theta})}_{0} r = 0$$

acceleration in radial direction ($\hat{e}_r$)

$$\vec{F} = F \hat{e}_r, \quad f = \frac{F}{m}$$

$$f = \frac{F}{m} = \ddot{r} - r\dot{\theta}^2$$

$$\text{We have } \vec{h} = \vec{r} \times \vec{v} = r^2 \dot{\theta} \hat{e}_3$$

$$h \hat{e}_3 = \vec{h}$$

$$h = r^2 \dot{\theta}$$

$$\dot{\theta} = \frac{h}{r^2}$$

$$f = \ddot{r} - \frac{h^2}{r^3} \quad (\text{eq 1}) \quad \text{Let } r = \frac{1}{u}, \quad u = \frac{1}{r}$$

$$\text{Note: } \frac{du}{dt} = \frac{du}{d\theta} \frac{d\theta}{dt}$$

$$\frac{dr}{dt} = \frac{-1}{u^2} \frac{du}{dt} \cdot \frac{d\theta}{dt} = \underbrace{-r^2 \dot{\theta}}_h \frac{du}{d\theta} = -h \frac{du}{d\theta}$$

$$\frac{d^2 r}{dt^2} = -h \frac{d}{dt} \left(\frac{du}{d\theta} \right) = -h \frac{d}{d\theta} \left(\frac{du}{d\theta} \right) \frac{d\theta}{dt} = -h \dot{\theta} \frac{d^2 u}{d\theta^2}$$

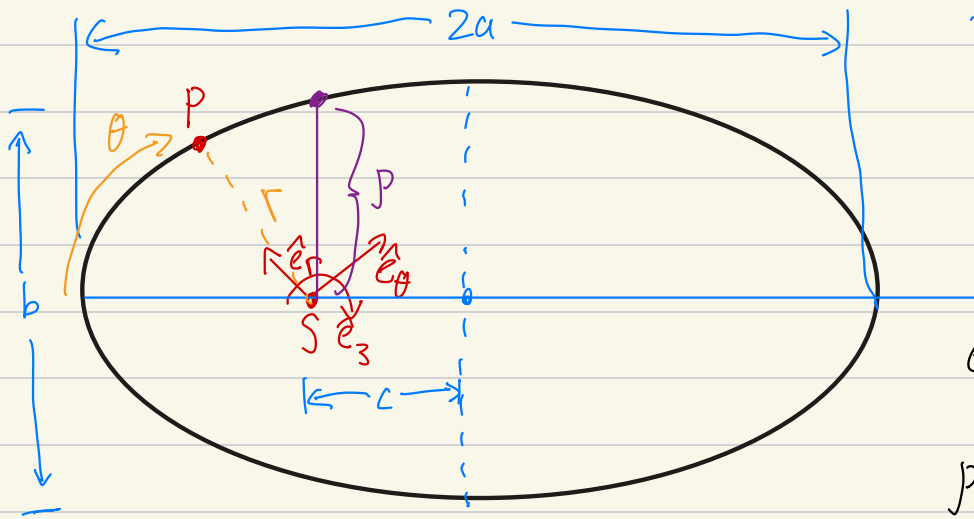
$$f = -h \dot{\theta} \frac{d^2 u}{d\theta^2} - h^2 u^3$$

$$h = r^2 \dot{\theta} = \text{const}$$

$$h = \frac{\dot{\theta}}{u^2}$$

$$\dot{\theta} = u^2 h$$

$$f = -h^2 u^2 \frac{du^2}{d\theta^2} - h^2 u^3 \quad (\text{Eq 2})$$



a = Semi-major axis length

c = foci distance ; $c^2 = a^2 - b^2$

e = eccentricity = $\frac{c}{a}$

p = Semi-latus rectum

$$p = a(1 - e^2) = \frac{b^2}{a}$$

$$\text{Area} = \pi ab$$

Equation of an ellipse in polar form:

$$r(\theta) = \frac{p}{1 + e \cos \theta}$$

$$u(\theta) = \frac{1 + e \cos \theta}{p} \quad \frac{du}{d\theta} = \frac{-e}{p} \sin \theta, \quad \frac{d^2 u}{d\theta^2} = \frac{-e}{p} \cos \theta$$

$$f = -h^2 u^2 \frac{du^2}{d\theta^2} - h^2 u^3 \rightarrow f = -h^2 u^2 \left[\frac{-e}{p} \cos \theta + \frac{1 + e \cos \theta}{p} \right] \rightarrow f = \frac{-h^2 u^2}{p} \rightarrow$$

$$f = \frac{-h^2}{p} \cdot \frac{1}{r^2} \quad \text{inverse square of distance}$$

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$$f = -\frac{h^2}{p} \cdot \frac{1}{r^2}$$

Kepler's 2nd law: $\frac{d\vec{A}}{dt} = \frac{1}{2}(\vec{r} \times \vec{v}) = \frac{\vec{h}}{2} \text{ const}$

$$\frac{dA}{dt} = \frac{h}{2}$$

Area of ellipse: $A = \int_0^{T(\text{planet year})} \frac{h}{2} dt = \frac{hT}{2}$

$$h = \frac{2A}{T}$$

$$A = \pi ab$$

$$p = \frac{b^2}{a}$$

$$h = \frac{2\pi ab}{T}$$

$$\frac{h^2}{p} = \left(\frac{2\pi ab}{T}\right)^2 \frac{a}{b^2} \rightarrow \frac{h^2}{p} = \frac{4\pi^2 a^3}{T^2} \quad (\text{eq 3})$$

Kepler's 3rd law: $T^2 \sim a^3$ where a is the mean radius of the ellipse

$\frac{a^3}{T^2}$ is the same for all the planets

$\frac{h^2}{p}$ is a property of Sun

$\frac{N}{kg} = \frac{kg \cdot m}{kg \cdot s^2} \quad \mu_0 = \frac{h^2}{p} \quad \text{Sun's gravitational parameter}$

$f = -\frac{\mu_0}{r^2}$ gravitational attraction force per unit mass from the Sun

$$\vec{f} = -\frac{\mu}{r^2} \hat{e}_r \quad \vec{f} = -\frac{\mu \vec{r}}{r^3} \quad \vec{F} = m\vec{f} = -\frac{\mu m}{r^2} \hat{e}_r$$

$$\ddot{\vec{r}} = -\frac{\mu \vec{r}}{r^3}$$

Newton: $\mu \sim M$ mass of primary

$\mu = GM$, $\mu_0 = GM_0$; G is universal gravitational constant

Value of G found in 1798 by Cavendish

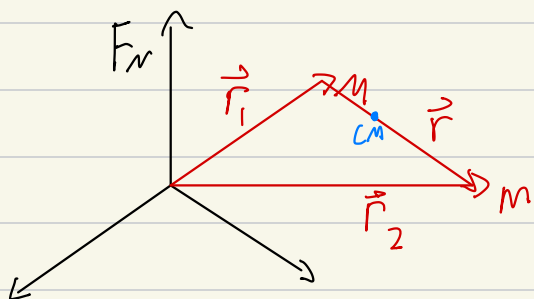
$$\vec{F} = \frac{-GMm}{r^2} \hat{e}_r \quad (\text{Law of universal gravitation}) \quad G = 6.674 \cdot 10^{-11} \frac{\text{m}^3}{\text{kg} \cdot \text{s}^2}$$

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$$F_{\text{Gravity}} = \frac{-GMm \hat{e}_r}{r^2}$$

2-body problem

M mass of object #1, m mass of object #2



$$\vec{F}_{\text{gravity}_2} = \frac{-GMm}{r^2} \cdot \frac{\vec{r}}{r}$$

$$\vec{F}_{\text{gravity}_1} = + \frac{GMm}{r^2} \cdot \frac{\vec{r}}{r}$$

$$m \ddot{\vec{r}}_2 = -\frac{GMm}{r^2} \left(\frac{\vec{r}}{r} \right) \quad M \ddot{\vec{r}}_1 = +\frac{GMm}{r^2} \left(\frac{\vec{r}}{r} \right)$$

Add the 2 eqns: $M \ddot{\vec{r}}_1 + m \ddot{\vec{r}}_2 = 0$

$$M \ddot{\vec{r}}_1 + m \ddot{\vec{r}}_2 = \overrightarrow{\text{Constant}} \rightarrow \text{Constant Vector}$$

$$\vec{r}_{CM} = \frac{M \vec{r}_1 + m \vec{r}_2}{M+m} \quad (\text{position of center of mass})$$

$$\ddot{\vec{r}}_{CM} = \frac{M \ddot{\vec{r}}_1 + m \ddot{\vec{r}}_2}{M+m} = \overrightarrow{\text{Constant}'} \quad (\text{different constant})$$

"Subtract" equations: (want to find $\ddot{\vec{r}} = \ddot{\vec{r}}_2 - \ddot{\vec{r}}_1$)

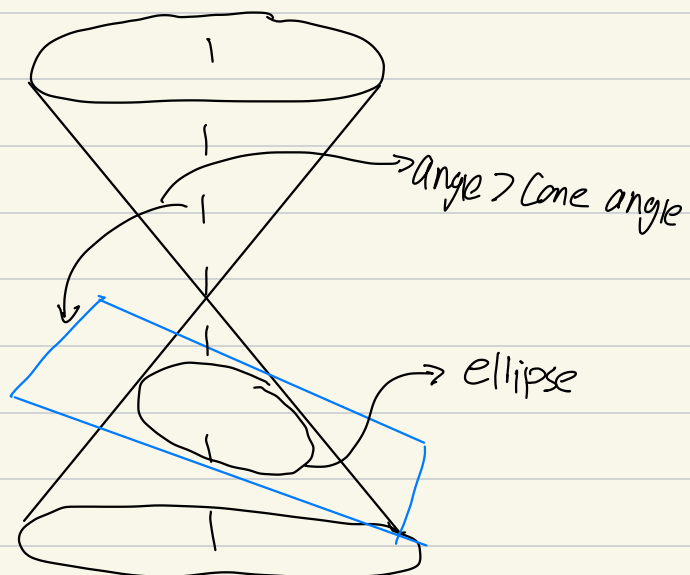
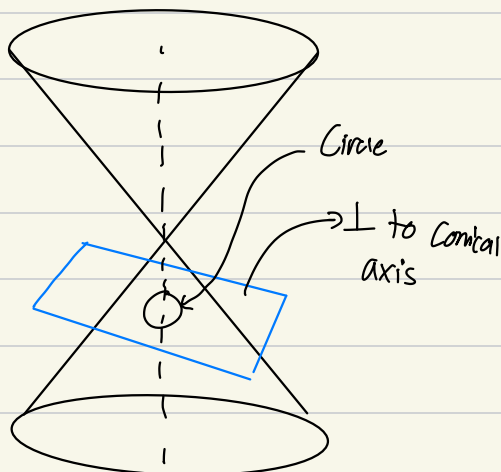
$$\ddot{\vec{r}} = -\frac{GM\vec{r}}{r^3} - \frac{Gm\vec{r}}{r^3} \rightarrow \ddot{\vec{r}} = -\frac{G(M+m)\vec{r}}{r^3}$$

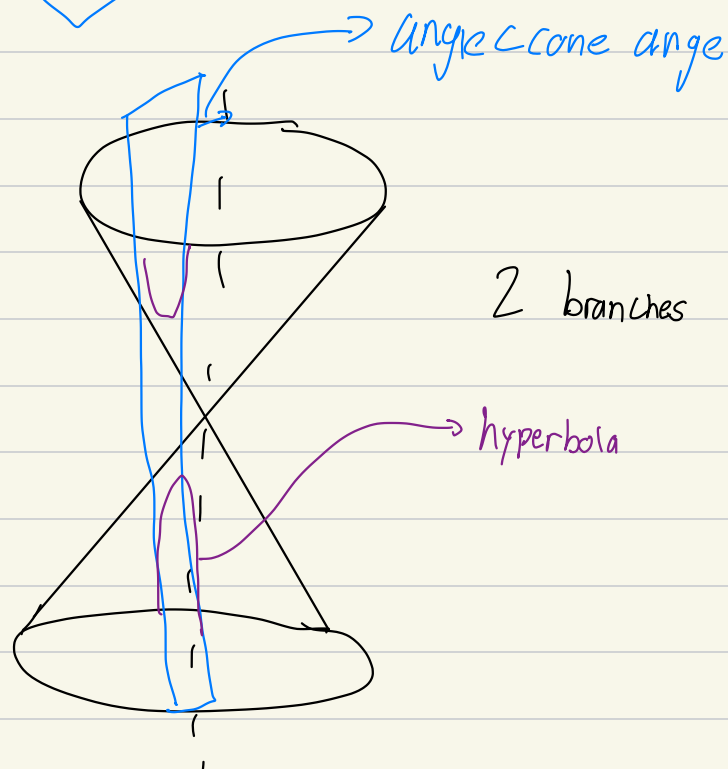
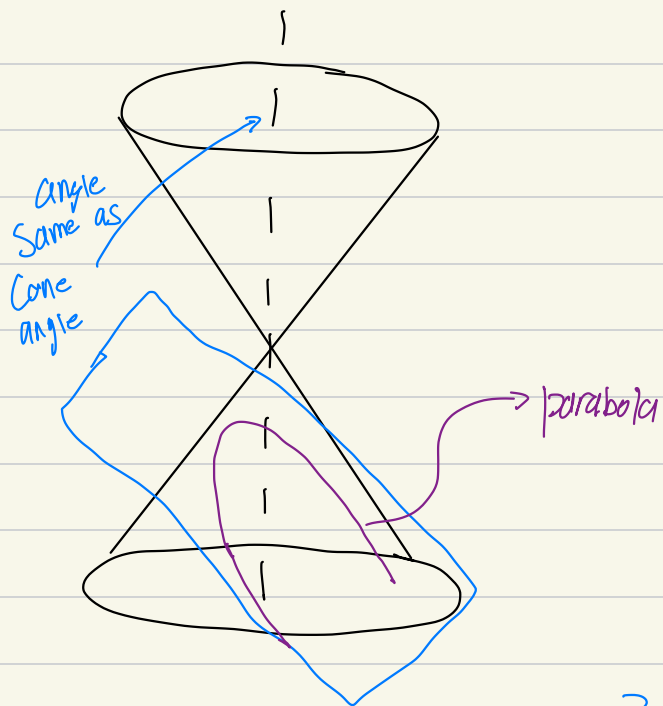
$$\ddot{\vec{r}} = -\mu \frac{\vec{r}}{r^2}$$

$\mu = G(M+m)$ for 2-body problem ; if $m \ll M$ then $\mu \approx GM$

Kepler - Ellipse

Newton - Ellipse + all other conic sections for 2 body problem





e - eccentricity

Circle $e=0$

ellipse $0 < e < 1$

parabola $e=1$

hyperbola $e > 1$

p Semi-latus rectum

